

IET, DAVV, Indore
B.E. II (IT-A & B)
Class Test III (Nov.-2022)
3 AIR C1: Discrete Mathematics

Time: 70 min

Max Marks: 20

Note: Attempt any **FOUR** questions. All questions carry equal marks. Write neatly and properly.

- Q.1. Using mathematical induction prove that $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$, $\forall n$. (05)
- Q.2. A parking lot has 31 visitors space, numbered from 0 to 30. The hashing function $h(x) = x \pmod{31}$ assigns spaces to cars according to the **first three digits** on a visitor's license plate. Which spaces are assigned by this hashing function to the cars that have these digits on their license plates: **31712, 91834, 00756, 10078, 11191, 31023, 53445, 21467, 76389**. Describe a procedure visitors should follow to find a free parking space, when the space they are assigned is occupied. (05)
- Q.3. Find the number of Mathematics students at a college taking atleast one of the language when 45 study German language, 65 study French language, 42 study Russian language, 20 French and German language, 25 French and Russian language, 15 German and Russian language and 8 study all the 3 languages. (05)
- Q.4. Construct the Hasse diagram for relation "a divides b" on the set $\{2, 3, 5, 10, 11, 15, 25\}$, explaining steps properly. (05)
- Q.5. Use Warshall's algorithm to find the transitive closure of the relation $R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 2), (3, 4), (4, 1)\}$ on the set $A = \{1, 2, 3, 4\}$. (05)
